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Module 1 Introduction

Lesson 1: Parameterized Priors and the Concept of Exchangeability

Reading: Parameterized Priors and the Concept of Exchangeability Readings
1h

Video: Parameterized Priors and Exchangeability Pt. 1
6 min

Video: Parameterized Priors and Exchangeability Pt. 2
6 min

Practice Assignment: Parameterized Priors and the Concept of Exchangeability Quiz
Submitted

Lesson 2: Analysis and Applications of Hierarchical Models

Lesson 3: Computational Techniques and Model Validation

Module 5 Summative Assessment

Module 5 Summary

Your grade: 100%

Your latest: 100% • Your highest score: 100% • To pass you need at least 80%. We keep your highest score.

Next item

Parameterized Priors and the Concept of Exchangeability Quiz

Instructions

Review Learning Objectives

1. What is a key feature of hierarchical models? **1 / 1 point**

   - They are only applicable to small datasets.
   - **They model data with hierarchical structures.**
   - They always use non-informative priors.
   - They assume all data points are independent.

   Correct
   Hierarchical models involve modeling data conditionally on certain parameters, which themselves have a probabilistic specification in terms of hyperparameters.

2. Why are simple nonhierarchical models usually inappropriate for hierarchical data? **1 / 1 point**

   - They are too complex for small datasets.
   - They do not require prior distributions.
   - **They tend to overfit or underfit data.**
   - They assume perfect data.

   Correct Like Dislike Report an issue
   Nonhierarchical models can either overfit or underfit data because they cannot appropriately balance the complexity of the model with the amount of data.

3. What is a major disadvantage of using a point estimate for α and β in the Beta distribution? **1 / 1 point**

   - It always leads to overfitting.
   - It requires a large sample size.
   - **It can ignore some posterior uncertainty.**
   - It is computationally intensive.

   Correct
   Using a point estimate for α and β ignores some posterior uncertainty.

Ready to practice Let's chat

Assignment details
Submitted Dec 6, 12:50 AM IST
Attempts Unlimited
Retry

Your grade
100%
To pass 80% or higher
View submission See feedback